

d) Bernoulli

$$E[N] = 7p + 20$$

$$\text{Var}(N) = 7^2 + \text{Var}(X) = 49p(1-p)$$

$$E[N^2] = 7^2(p-p^2) + (7p+20)^2$$

$$= 49(p-p^2) + (7p+20)^2$$

$$= 49p - 49p^2 + 49p^2 + 400 + 280p$$

$$= 329p + 400$$